BAGSENSE · FYP1 · BICS 4321

Three commercial products, and what they leave open

Apple AirTag · Impinj Speedway R420 · BAGTAG Flex

Compiled 29 October 2026 · prices in RM at ECB reference rates

Ahmad Adam Hakimi (2418711) · Amru Hazem (2419467)

Why these three, and not three trackers

Three products on the same axis would prove nothing. One per axis shows the landscape is understood.

AXIS 1 · THE BASELINE

Apple AirTag

RM129 per tag

What people buy today.

The product our problem statement argues against.

AXIS 2 · SAME PHYSICS, OTHER SCALE

Impinj Speedway R420

RM7,642 for the reader

Passive UHF inventory, proven in industry — but fixed, mains-powered, not carried.

AXIS 3 · SAME FORM FACTOR

BAGTAG Flex

RM336 per device

A display on a bag, batteryless — yet it knows nothing about the contents.

Prices: vendor published prices, accessed 29 Oct 2026. GBP and EUR converted at ECB reference rates of 28 Oct 2026.

AirTag 2nd generation

Apple Inc. · the baseline BagSense argues against

What it does	Publishes its location to the Find My network: map, sound, and UWB Precision Finding when close.
What it does not do	It does not know what is in the bag. One tag tracks one item, and each item needs its own tag.
Power	One CR2032 coin cell per tag (user-replaceable). Every protected item carries a battery.
Ecosystem	Requires an iPhone on iOS 26 or later, plus the Find My network.
Cost to protect 10 items	RM1,290 at list price — three four-packs (12 tags) would be RM1,287.

PRICE, APPLE MALAYSIA

RM129

single · RM429 for a four-pack

Why it is here: this is the cost and battery argument in our proposal — now with a sourced price.

Source: apple.com/my/airtag, apple.com/my/shop/buy-airtag, Apple newsroom 26 Jan 2026, Apple Support tech specs. Accessed 29 Oct 2026.

Impinj Speedway R420

Impinj · the same physics, sold as infrastructure

PRICE, READER ONLY

RM7,642

£1,410.79 ex VAT, GX1 variant (covers Malaysia)

What it does	Interrogates passive UHF RAIN tags in range and reports their EPC identifiers — a genuine inventory of what is inside the read zone.
Scale	Four antenna ports, PoE or mains. Built for shelves, doors, cabinets and conveyors.
Not included	Antennas from about RM1,440 each, power supply, cabling, and the inventory software.
Why it strengthens our case	It proves passive UHF item-level inventory works commercially. BagSense does not invent the technology — it moves it into a bag that a person carries.

Same band as our design: the GX1 variant is the one specified for Malaysia.

Source: rfid-uk.com product page (Impinj Speedway R420), accessed 29 Oct 2026. RM converted at 1 GBP = RM5.417.

BAGTAG Flex

BAGTAG B.V., Netherlands · the closest thing to a smart bag

PRICE, DIRECT

RM336

€74 on the product page (an earlier listing showed €89)

What it does	Shows a digital bag tag on an e-ink panel, written from the phone over NFC, so a bag can be checked in without a paper label.
The interesting part	Batteryless, and the e-ink panel keeps its image with no power — the same display argument our design uses.
What it does not do	No sensing, no inventory, no radio for reading anything. It replaces a label, it does not know what the bag holds.
Ecosystem caveat	Only works with airlines on the BAGTAG platform — check the supported-airlines list for Malaysia before treating it as available here.

Same form factor, different job — and the panel idea is already commercially proven.

Source: bagtag.com/product/bagtag-flex/, accessed 29 Oct 2026. RM converted at 1 EUR = RM4.539.

Side by side

The table for Chapter 2. BagSense is included so the comparison is anchored, not implied.

	AirTag	Impinj R420	BAGTAG Flex	BagSense
Question it answers	Where is this item?	What is in this read zone?	What does the bag tag say?	**What is in this bag?**
Knows bag contents	No	Yes — in its zone	No	**Yes**
Cost to cover 10 items	RM1,290	N/A — zone, not items	N/A	**RM5.60 for 10 labels**
Power per item	CR2032 per tag	Mains / PoE	None	**One pack, whole bag**
Needs a phone	Yes	No	Yes	**No (status on bag)**
Glanceable on the object	No	No	Yes	**Yes**
Carried by the user	Tag only	No	Yes	**Yes**
Cost to start	RM129	RM7,642 + antennas	RM336	**RM571 prototype**

Vendor prices and FX accessed 29 Oct 2026. BagSense costs from the project BOM (RM571) and quoted label price (RM0.56).

What this changes in the report

The gap holds

No product in this set reports the contents of a bag a person carries. AirTag answers “where”, BAGTAG answers “what label”, the Impinj reader answers “what is in this zone” — but sits on a wall.

The cost claim is now sourced

“RM120–RM160” in the proposal was inherited, not verified. AirTag is RM129 and SmartTag2 RRP is RM149 — both now cited with a date, and ten items is RM1,290, not “over RM1,200”.

The technology is not the novelty

Passive UHF inventory is commercially mature. BagSense contributes the constraint set — a small cavity, fabric, a battery, and a display on the bag — not a new radio. Say that in Chapter 2 and the contribution sounds considered rather than overclaimed.

Proposal wording updated to match: “cost between RM120–RM160” → “RM129 (Apple Malaysia) and RM149 RRP (Samsung Malaysia)”.

Sources and caveats

- Apple AirTag (MY) — apple.com/my/airtag/ and apple.com/my/shop/buy-airtag · accessed 29 Oct 2026
- Apple newsroom, 26 Jan 2026 — RM129 single / RM429 four-pack
- Apple Support — AirTag (2nd gen) tech specs: user-replaceable CR2032, IP67
- Samsung Malaysia published accessory price list — Galaxy SmartTag2 RRP RM149
- RFID-UK — Impinj Speedway R420, £1,410.79 ex VAT (GX1 variant, covers Malaysia)
- BAGTAG — bagtag.com/product/bagtag-flex/ — €74, batteryless, NFC e-ink
- Aenon Solutions, Penang — local RFID smart cabinets, price on request (not profiled)
- FX: frankfurter.dev (ECB reference rates), 28 Oct 2026 — EUR 4.539, GBP 5.417, USD 4.077 per RM

Caveats: cross-border prices exclude shipping, duty and SST, so the landed cost is higher than shown. Vendor specifications are vendor claims, not independent measurements. Marketplace prices are excluded — only list prices from the vendor are used, because a discount is not a specification.